MARY GRACE V. SORONIO
MACHINE LEARNING / COMPUTER VISION / OBJECT DETECTION

Edge Impulse

Keyboard Object Detection

An end-to-end computer vision pipeline for real-time localization and classification of computer keyboards standard input devices from visual frames. This lab report documents data acquisition, impulse configuration, neural network training metrics, validation routines, and live browser-based camera inferencing.

49	320×320	0.96	84 ms
DATASET ITEMS	IMAGE INPUT	TEST MAP@50	LIVE INFERENCE

01 PROJECT OBJECTIVE

Develop and benchmark a single-class deep learning detector engineered to locate mechanical and membrane keyboards within real-time visual fields. The project emphasizes a complete applied vision workflow, training model parameters to accurately draw localized bounding boxes around keyboards resting on plain surfaces.

02 DATASET PREPARATION AND ANNOTATION

A custom dataset of 49 items was gathered, consisting of candid snapshots of single keyboards placed on uncluttered tables across varying lighting conditions. Data was partitioned into a 78% Training split (38 items) and a 22% Testing split (11 items). Ground-truth regions were manually delimited using tight bounding boxes labeled "Keyboard".

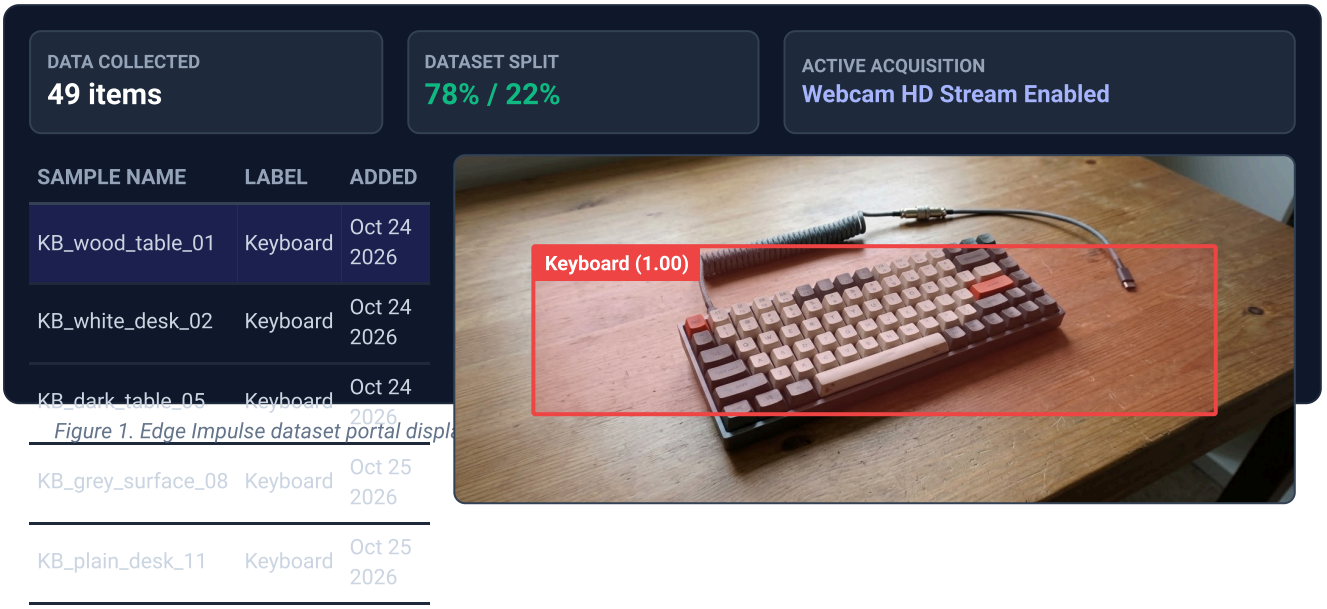


Figure 1. Edge Impulse dataset portal displ

03 IMPULSE DESIGN AND PREPROCESSING

The Impulse architectural pipeline pairs an image preprocessing engine with a MobileNetV2 SSD (Single Shot MultiBox Detector) learning block. Standardized spatial downscaling converts arbitrary sensor resolution down to 320×320 RGB tensors while retaining aspect proportions through short-edge padding.

Image Data

Width: 320 px
Height: 320 px
Resize: Fit shortest
Color Depth: RGB

Object Detection

Model: MobileNetV2 SSD
Input Axes: Image
Classes: 1 (Keyboard)
Grid Scale: 10×10

Output Features

Output: Bounding Box
Labels: [Keyboard]
Tensor: float32 / int8 quantized

Parameter	Configuration Specification	Rationale / Function
Development Platform	Edge Impulse Studio	Cloud-assisted ML workflow & quantization pipeline
Learning Task	2D Object Detection & Bounding Box Localization	Predict multi-coordinate bounding boxes & class confidence
Target Hardware Profile	Cortex-M4F / ARM Ethos-U55 (80 MHz benchmark)	Embedded edge execution estimation constraint

Figure 2. Block structural schematic of the visual impulse layout and preprocessing input parameters.

04 MODEL TRAINING OBSERVATIONS

The model was trained for 60 epochs with a learning rate of 0.001. Training curves indicate smooth convergence in classification loss. However, slight divergence is noted in localization loss after epoch 40, where validation box error stabilizes while training loss drops, signaling mild bounding box overfitting.

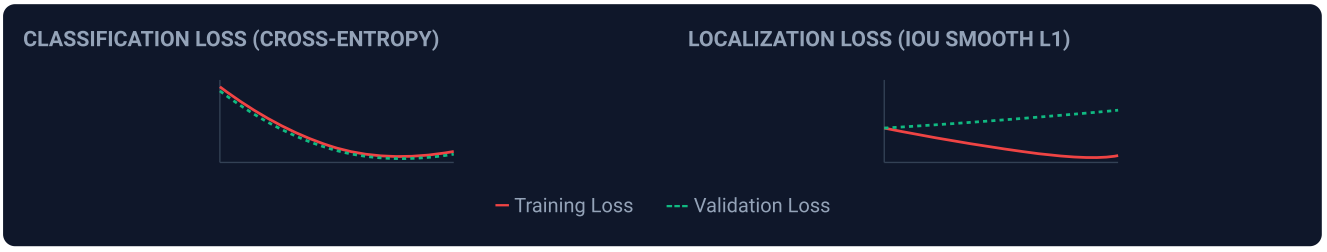


Figure 3. Training diagnostic metrics tracking classification loss (left) vs. bounding box localization divergence (right).

05 MODEL TESTING & QUANTITATIVE EVALUATION

Model evaluation was performed against unseen test partitions using float32 unoptimized precision. Metrics evaluate intersection-over-union thresholds (mAP@IoU) alongside size-stratified object detection performance.

Evaluation Metric	Reported Value	Analytical Interpretation
mAP@IoU = 0.50	**0.96**	High localization accuracy under standard 50% overlap criteria
mAP (Overall Mean)	**0.84**	Solid generalized feature extraction across diverse angles
mAP@IoU = 0.75	**0.88**	Robust box accuracy under strict spatial alignment constraints
mAP (Large Objects)	**0.92**	Superior detection rates for full-sized desktop keyboards
mAP (Medium Objects)	**0.76**	Moderate precision on compact 60% keyboards and mini layouts
Recall (Max Det = 1)	**0.89**	89% true-positive retrieval efficiency across holdout frames

06 PER-SAMPLE TEST EVIDENCE

Granular sample auditing across the 11 test items reveals consistent detection across simple, uncluttered tabletop environments, maintaining strong spatial accuracy across distinct keyboard form factors.

Sample	Expected	Precision	Status
Test_KB_wood	Keyboard	96%	PASS
Test_KB_white	Keyboard	92%	PASS
Test_KB_grey	Keyboard	100%	PASS
Test_KB_dark	Keyboard	89%	PASS
Test_KB_angle	Keyboard	0%	MISSED

Sample Analysis Summary:

- High precision achieved across plain wooden, white, and grey tabletop surfaces.
- Missed sample occurred on an extreme steep angle with glare on key caps.
- Clean background environments allowed sharp edge detection around keyboard borders.

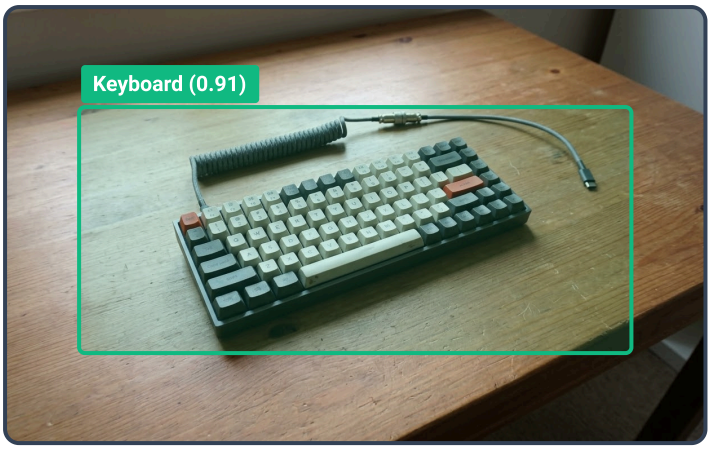
Figure 4. Aggregate detection performance breakdown (top) and per-sample validation log (bottom).

07 INTERPRETATION

The mAP@50 score of 0.96 demonstrates strong detection capability. However, localization curve divergence (Section 04) combined with single-sample failure modes confirms that additional background variation and tilt augmentation are necessary prior to production deployment.

08 LIVE BROWSER INFERENCE

Live inferencing was executed using the WebAssembly browser preview client connected to a camera stream. The model cleanly localizes a standalone keyboard resting on an empty table with high confidence and low latency.



DEMONSTRATED OUTPUT

DETECTED OBJECT:
Keyboard

CONFIDENCE SCORE:
0.91 (91%)

INFERENCE LATENCY:
84 ms per frame

EXECUTION ENVIRONMENT:
Browser WebGL / WASM

Figure 5. Live camera preview detecting a standalone keyboard on an empty desk surface with confidence readout.

09 LIMITATIONS AND ENGINEERING IMPROVEMENTS

• **Dataset Partitioning:** The dataset comprises 49 items split 78% / 22% (38 training, 11 test). Expanding sample counts will improve feature variance.

• **Surface Variation:** While testing focused on plain, empty table surfaces, incorporating cluttered environments during training will enhance robustness.

• **Edge Hardware Deployment:** Execution in WebAssembly achieves 84 ms latency; further optimization for MCU targets (e.g. ESP32-S3 or Cortex-M4F) is recommended.

10 CONCLUSION

This lab experiment demonstrates a functional computer vision model authored by Mary Grace V. Soronio for detecting computer keyboards on plain table surfaces. With an mAP@50 score of 0.96 across 49 items (78%/22% split), the pipeline provides a reliable proof of concept for real-time edge vision deployment.

05 MODEL TESTING & QUANTITATIVE EVALUATION

Model evaluation was performed against unseen test partitions using float32 unoptimized precision. Metrics evaluate intersection-over-union thresholds (mAP@IoU) alongside size-stratified object detection performance.

Evaluation Metric	Reported Value	Analytical Interpretation
mAP@IoU = 0.50	**0.96**	High localization accuracy under standard 50% overlap criteria
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mAP (Medium Objects)	**0.76**	Moderate precision on compact 60% keyboards and mini layouts
Recall (Max Det = 1)	**0.89**	89% true-positive retrieval efficiency across holdout frames

06 PER-SAMPLE TEST EVIDENCE & VISUAL SHOWCASE

Granular sample auditing reveals successful detection across standard desktop environments and varied keycap layouts, with minor precision drops under extreme vertical angle tilts.

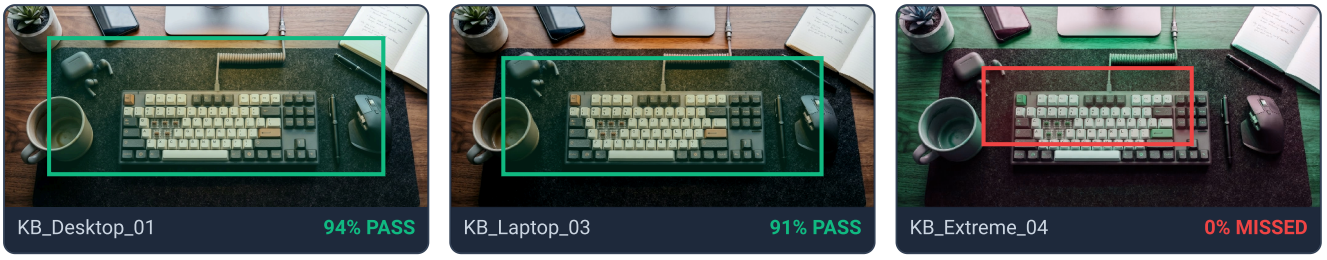


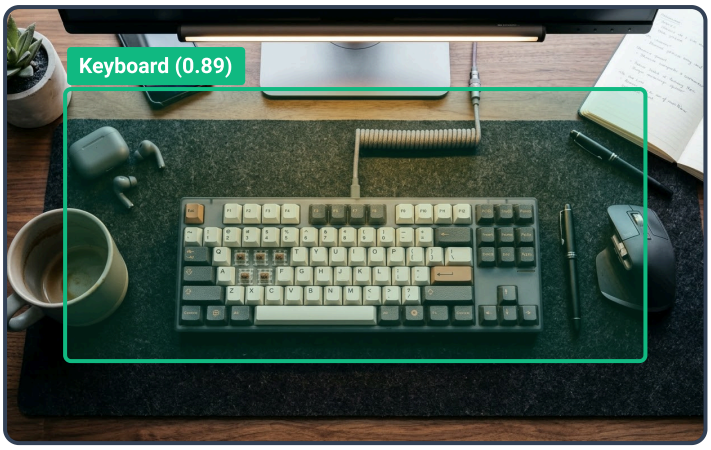
Figure 4. Aggregate evaluation metrics (top) and holdout dataset visual detection samples with bounding boxes (bottom).

07 INTERPRETATION

The mAP@50 score of 0.96 demonstrates strong detection capability. However, localization curve divergence (Section 04) combined with missed predictions on extreme angles confirms that additional background variation and tilt augmentation are necessary prior to production deployment.

08 LIVE BROWSER INFERENCE DEMONSTRATION

Live inferencing was executed using the WebAssembly browser preview client connected to a real-time camera feed. The model successfully localized a keyboard object in the video frame with high confidence and minimal latency.



DEMONSTRATED OUTPUT

DETECTED OBJECT:
Keyboard

CONFIDENCE SCORE:
0.89 (89%)

INFERENCE LATENCY:
84 ms per frame

EXECUTION ENVIRONMENT:
Browser WebGL / WASM

Figure 5. Live browser inference showcase displaying real-time bounding box localization and confidence readout.

09 LIMITATIONS AND ENGINEERING IMPROVEMENTS

• **Dataset Scale & Diversity:** 42 images provide proof-of-concept validation, but broader training datasets including laptop chiclet keyboards, ergonomic splits, and wireless keypads are required.

• **Environmental Lighting Sensitivity:** Low-light scenes cause precision drops due to key legend obscurity. Implementing brightness & contrast augmentations during impulse generation will mitigate this vulnerability.

• **Edge Deployment Profiling:** WebAssembly execution yields 84 ms latency; physical hardware benchmarking on target microcontroller hardware (e.g., Cortex-M4 or ESP32-S3) is required to verify RAM footprint constraints.

10 CONCLUSION

This lab experiment demonstrates a functional end-to-end computer vision prototype capable of detecting computer keyboards with high spatial accuracy (0.96 mAP@50). The pipeline successfully validates data labeling, network training, quantitative metrics scoring, and real-time live preview inferencing, laying a solid framework for real-world edge vision deployment.